

①

$$L = \{a^i b^j c^k \mid i = j \text{ o } j \leq k\}$$

$$S \rightarrow AE \mid B$$

$$A \rightarrow aAb \mid \lambda \quad \leftarrow N(a) = N(b)$$

$$B \rightarrow CD$$

$$C \rightarrow aC \mid \lambda \quad \alpha = aA$$

$$D \rightarrow bDc \mid E$$

$$E \rightarrow cE \mid \lambda$$

$$AE = N(a) = N(b) \quad \uparrow \text{minúscula}$$

$$B = N(c) \geq N(b)$$

Paso 1: Eliminar producciones unitarias

- Localizar símbolos anulables

$$A_0 = \{E, C, A\}$$

$$A_1 = \{E, C, D, A\}$$

$$A_2 = \{A, B, C, D, E\}$$

$$A_3 = \{S, A, B, C, D, E\}$$

Reescribimos las normas

$$S \rightarrow A \mid E \mid AE \mid B \mid \lambda \quad \leftarrow \lambda \in L$$

$$A \rightarrow ab \mid aAb$$

$$B \rightarrow C \mid D \mid CD$$

$$C \rightarrow a \mid aC \mid \lambda$$

$$D \rightarrow bc \mid bDc \mid E$$

$$E \rightarrow c \mid \lambda \mid cE \quad \uparrow \text{absurdo}$$

Paso 3: Eliminar símbolos inútiles

1º Generadores:

$$G_0 = \{a, b, c\}$$

$$G_1 = \{A, C, D, E, a, b\}$$

$$G_2 = \{A, B, C, D, E, a, b, c\}$$

2º Alcanzables

$$A_0 = \{S\}$$

$$A_1 = \{S, A, C, D, E\}$$

$$A_2 = \{S, A, C, D, E\}$$

Símbolo inútil = B

Reescribo normas:

$$S \rightarrow ab \mid aAb \mid a \mid aC \mid bc \mid bDc \mid c \mid cE \mid CD \mid AE$$

$$A \rightarrow ab \mid aAb$$

$$C \rightarrow a \mid aC$$

$$D \rightarrow bc \mid bDc \mid c \mid cE$$

$$E \rightarrow c \mid cE$$

Paso 2: Identificar pares unitarios:

$$\checkmark (A, A)$$

$$\checkmark (B, B) \quad (B, C) \quad (B, D) \quad (B, E)$$

$$\checkmark (C, C)$$

$$\checkmark (D, D) \quad (D, E)$$

$$\checkmark (E, E)$$

$$(S, S) \quad (S, A) \quad (S, E) \quad (S, B) \quad (S, D)$$

Reescribimos normas:

$$S \rightarrow ab \mid aAb \mid a \mid aC \mid bc \mid bDc \mid c \mid cE \mid CD \mid AE$$

$$A \rightarrow ab \mid aAb$$

$$B \rightarrow a \mid aC \mid bc \mid bDc \mid c \mid cE \mid CD$$

$$C \rightarrow a \mid aC$$

$$D \rightarrow bc \mid bDc \mid c \mid cE$$

$$E \rightarrow c \mid cE$$